


# DATA ANALYTICS

## UE23CS342AA2

### UNIT-1

**Lecture 8: Data preprocessing - Transformations and Discretization - 1**

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# Data Analytics

## Unit 1

### Lecture 8: Data preprocessing - Transformations and Discretization - 1

**Slides excerpted from:** Data Mining : Concepts and Techniques by Han, Kamber and Pei, 3rd Edition

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## Data Transformation

A function that maps the entire set of values of a given attribute to a new set of replacement values such that each old value can be identified with one of the new values.

Methods

- Smoothing
- Attribute construction
- Aggregation
- Normalization
- Discretization
- Concept hierarchy generation

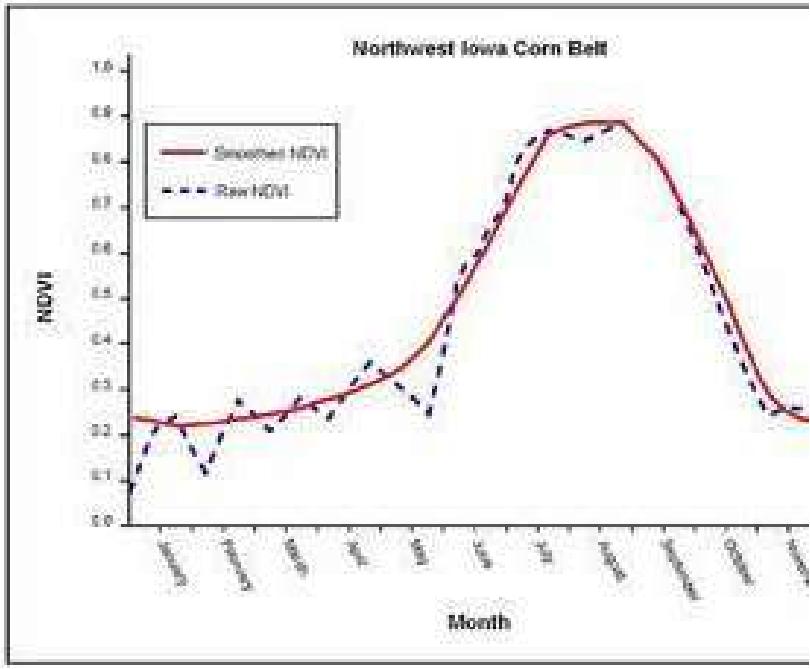
# DATA ANALYTICS

## Smoothing

Removal of noise in data.

Techniques:

- Binning
- Regression
- Clustering
- Simple average – Time series data
- Weighted average – Time series data
- Exponential smoothening – Time series data
- Gaussian filter - Image



# DATA ANALYTICS

## Normalization

Need for normalization :

- The unit of measurement can affect data analysis.
- For example , changing the unit of measurement of height from inches to cm can lead to different results.
- An attribute in smaller units results in a larger range. This attribute will be given a higher weight(preference) by few distance-based algorithms like KNN(K-nearest neighbors).
- For example , income of a person is generally a few orders higher than their height.
- To overcome this , data is generally transformed to a range such as [-1,1] or [0,1].

- Normalization by Decimal Scaling : A number *v* is normalized by $v' = \frac{v}{10^j}$

Where *j* is the smallest integer such that Max(|v'|)<1
For example , [-616,973] is normalized to [-0.616,0.973] where *j*=3.

## Normalization

- Min-Max Normalization : Performs a linear transformation. It transforms the values from [$min_A$, $max_A$] to [new_$min_A$, new_$max_A$]. A value *v* is transformed by

$$v' = \frac{v - min_A}{max_A - min_A}(new_max_A - new_min_A) + new_min_A$$

- Example : Let the income range \$12,000 to \$98,000 be normalized to [0.0,1.0]. Find out the mapping for \$73,600

$$\frac{73{,}600 - 12{,}000}{98{,}000 - 12{,}000}(1.0 - 0) + 0 = 0.716$$

- This preserves the relationship among the original data values. It will encounter an “out-of-bounds” error if an input which is outside the range of original data is provided.

## Normalization

- Z-score normalization : The values for an attribute A are normalized based on the mean and standard deviation of A. A value *v* can be normalized by

$$v' = \frac{v - \mu_A}{\sigma_A}$$

Where $\mu_A$ is the mean and $\sigma_A$ the standard deviation.

- Example : Let μ= 54,000 and σ = 16,000. Then z-score of 73,600 is

$$\frac{73{,}600 - 54{,}000}{16{,}000} = 1.225$$

- This method of normalization is useful when the actual minimum or maximum value of A is unknown or there are outliers in A.
- A variation of this is obtained by using Mean Absolute Deviation (MAD) instead of standard deviation. It is less susceptible to outliers.

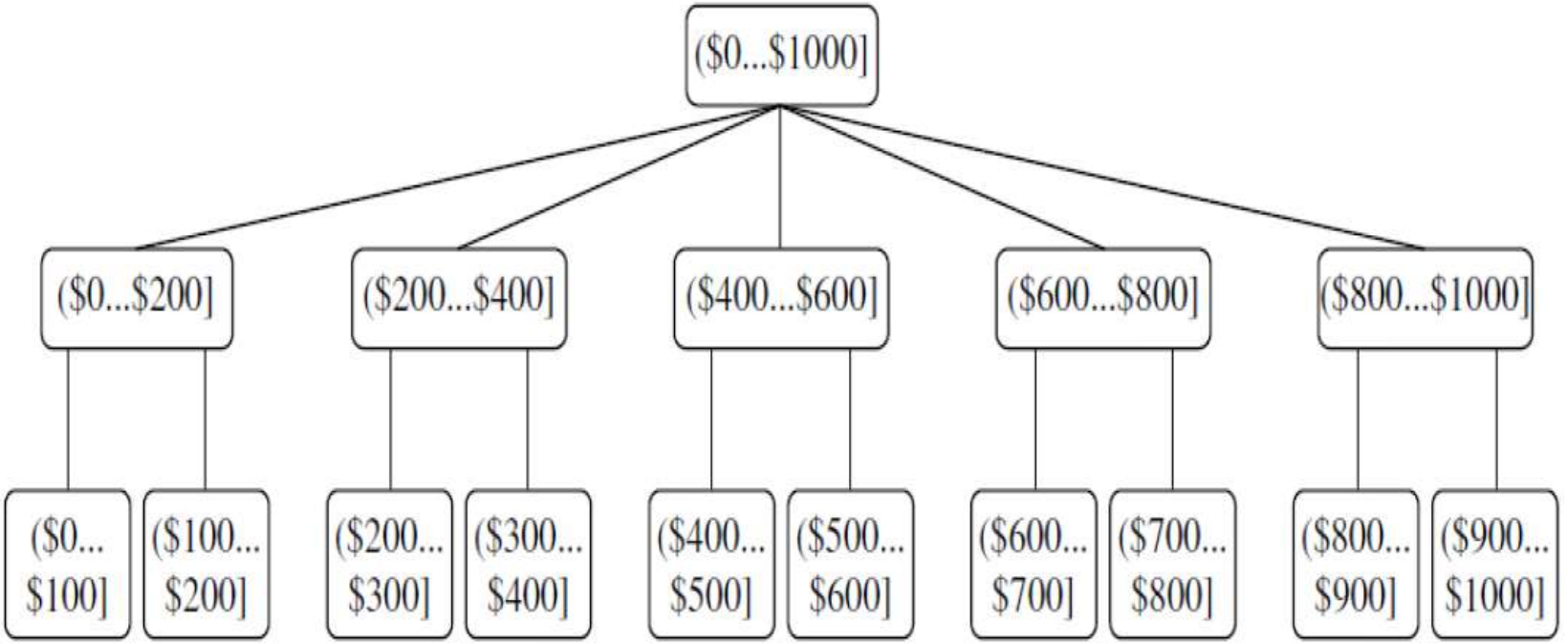
## Concept Hierarchy Generation

- Concept hierarchy organizes concepts ( Attribute values ) hierarchically by representing a series of mappings from a set of low-level concepts to a high-level , generalized concepts.
- It facilitates drilling and rolling in data warehouses to view data in multiple granularity.
- Method : Recursively reduce the data by collecting and replacing low level concepts (such as numeric values for age) by higher level concepts ( such as kids, teenagers, adults, senior citizens).
- They can be explicitly specified by domain experts and/or data warehouse designers.

## Concept Hierarchy Generation for Numeric Data

Discretization methods discussed till now can be used for numeric data.

Example:



## Concept Hierarchy Generation for Nominal Data

1) Specification of a partial ordering of attributes explicitly at the schema level
   - A user or an expert defines a concept hierarchy by specifying a partial or a total ordering of attributes at the schema level.
   - For example , suppose a relational database contains the attributes *street, city, state* and *country* . *Location* dimension of the data warehouse may contain the same attributes.
   - A hierarchy can be defined by specifying the total ordering among these attributes at the schema level

$$street < city < state < country$$

## Concept Hierarchy Generation for Nominal Data

2) Specification of a portion of hierarchy by explicit data grouping

- A portion of the concept hierarchy is manually defined.
- In a large database , it is unrealistic to define the entire concept hierarchy by explicit value enumeration.
- However , we can easily specify explicit groupings for a small portion of intermediate-level data.
- For example , after specifying state and country form a hierarchy at the schema level , a user can define few intermediate levels manually

{Karnataka , Tamil Nadu, Kerala, Andhra Pradesh, Telangana} < South India

## Concept Hierarchy Generation for Nominal Data

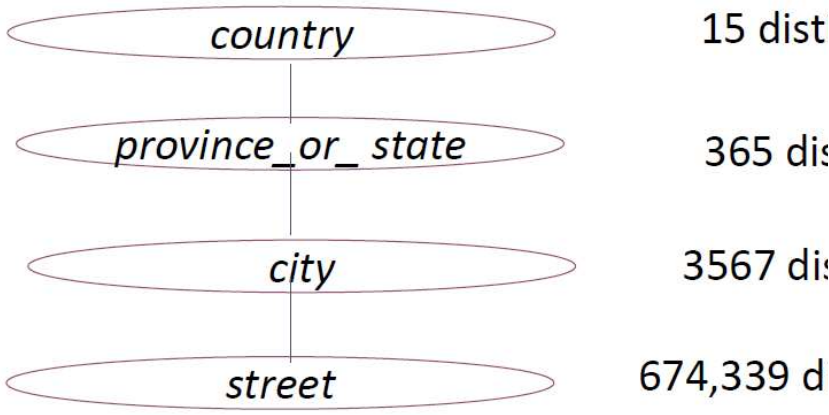
3) Specification of only a partial set of attributes.

- At times , a user can have a vague idea about what should be included in the hierarchy.
- The user may have included only a small subset of the relevant attributes in the hierarchy specification.
- For example, instead of including all the hierarchically relevant attributes for *location* , the user might have specified only *street* and *city*.
- To handle this , embed data semantics into the database schema. Hence one attribute will trigger a whole group of linked attributes to be added to the hierarchy. For example , when *city* is added , it would automatically include *state* and *country* as they are semantically related.

## Concept Hierarchy Generation for Nominal Data

### 4) Automatic generation of hierarchies by analysis of distinct values per attribute

- Few hierarchies can be automatically generated based on the analysis of the number of distinct values per attribute in the dataset.
- The attribute with the most distinct values is placed at the lowest level of the hierarchy.




Note : This method is not foolproof. For example, a time dimension in a databas contain 20 distinct *years* , 12 distinct *months* and 7 distinct *days of the week.* Howev doesn't suggest that the time hierarchy should be *year<month<days of the week* .

# DATA ANALYTICS

## Test your understanding!

- Which method of normalization must one choose if they are dealing with a lot of outliers and don't know the range of their data?

Solution

**Z-Score Normalization**

- Which split-point is preferred for discretization using decision trees?

Solution

**A Split point which results in least entropy.**

- Which normalization method strictly works in the range of input data?

Solution

**Min-Max Normalization**

# DATA ANALYTICS

## Test your understanding!

- Consider a set of Unsorted data for price in dollars
  8 16, 9, 15, 21, 21, 24, 30, 26, 27, 30, 34

1)Smooth the data by equal frequency bins
2)On the results of part (1) apply smoothing by bin means

Solution:

1)

**Bin 1:** 8, 9, 15, 16

**Bin 2:** 21, 21, 24, 26,

**Bin 3:** 27, 30, 30, 34

2)

**For Bin 1:**

(8+ 9 + 15 +16 / 4) = 12

(4 indicating the total values like 8, 9 , 15, 16)

Bin 1 = 12, 12, 12, 12

**For Bin 2:**

(21 + 21 + 24 + 26 / 4) = 23

Bin 2 = 23, 23, 23, 23

**For Bin 3:**

(27 + 30 + 30 + 34 / 4) = 30

Bin 3 = 30, 30, 30, 30

## References


- **Data Mining: Concepts and Techniques** by Jiawei Han, Micheline Kamber and Jian Pei, The Morgan Kaufmann Series in Data Management Systems, 3rd Edition Chapter 3.5
- https://t4tutorials.com/binning-methods-for-data-smoothing-in-data-mining/




# THANK YOU

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